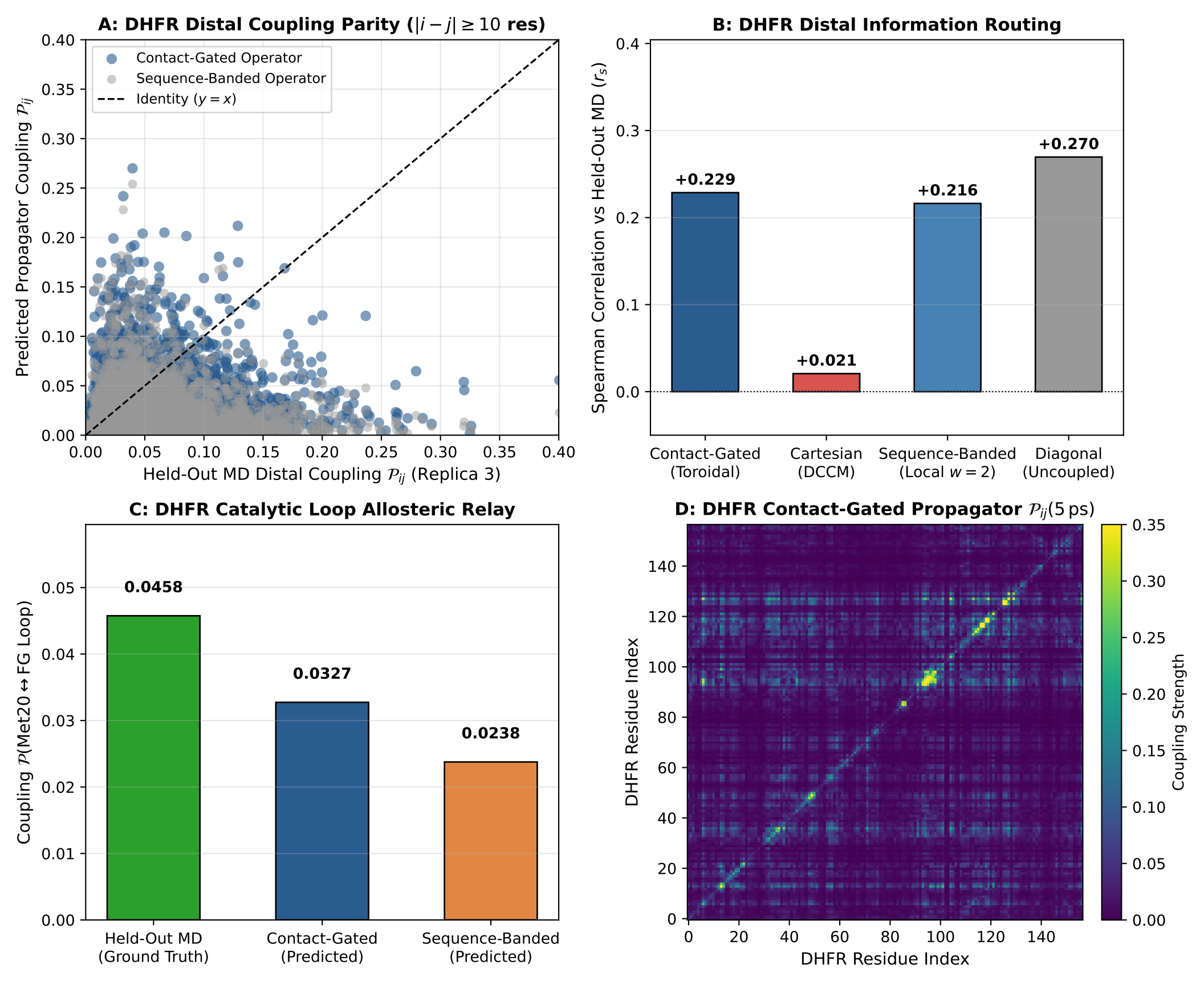

A: DHFR Distal Coupling Parity ($|i-j| \geq 10$ res)
Contact-Gated Operator
Sequence-Banded Operator
Identity ($y = x$)
Predicted Propagator Coupling $\mathcal{P}_{ij}$
Held-Out MD Distal Coupling $\mathcal{P}_{ij}$ (Replica 3)
B: DHFR Distal Information Routing
Spearman Correlation vs Held-Out MD (r_s)
+0.229
+0.021
+0.216
+0.270
Contact-Gated (Toroidal)
Cartesian (DCCM)
Sequence-Banded (Local $w = 2$)
Diagonal (Uncoupled)
C: DHFR Catalytic Loop Allosteric Relay
Coupling $\mathcal{P}$(Met20 ↔ FG Loop)
0.0458
0.0327
0.0238
Held-Out MD (Ground Truth)
Contact-Gated (Predicted)
Sequence-Banded (Predicted)
D: DHFR Contact-Gated Propagator $\mathcal{P}_{ij}$(5 ps)
DHFR Residue Index
DHFR Residue Index
Coupling Strength